

Research and developments on the eastern Bering Sea walleye pollock stock assessment

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1 Summary

The work presented below was based on the same data sets used in the 2024 assessment. We broke the work into two parts, first is the discussion paper presented to the Center for Independent Experts (CIE) review panel prepared in May 2025 ([link here](#)).

Here's a clean summary you can drop near the top of Part 2 (or use as an executive summary box). I kept it tight but complete, mirroring the structure of your qmd.

Summary This document is a follow-up to the May 2025 CIE review of the Eastern Bering Sea (EBS) walleye pollock stock assessment. The review panel provided a set of recommendations to enhance the model structure, address key uncertainties, and improve transparency. This two-part response addresses those recommendations in sequence. Part 1 (May 2025) is an updated version of a discussion paper presented to the review panel. findings and outlined a roadmap for addressing them within the existing ADMB framework, focusing on Russian removals, process error, selectivity, and growth. This document, Part

Part 2 documents actions taken since the May 2025 CIE review and provides analyses that inform the September 2025 NPFMC review. It focuses on (i) migrating the operational model from ADMB to RTMB, (ii) bounding the influence of Russian removals, (iii) process-error and likelihood refinements, and (iv) cross-basin comparisons.

Key outcomes since the CIE review

- RTMB transition (Completed). The ADMB assessment was ported to RTMB and verified:
- RTMB reproduces ADMB outputs for SSB and age-1 recruitment with negligible differences.
- Enables random-walk selectivity/recruitment, marginal-likelihood error propagation, and RTMB flexibility to explore random effects.
- Russian removals (In progress). Scenario modeling underway using Russian catch assumptions of ~20–45% of U.S. catch (per reviewer guidance and Sharov et al.). The goal is to bound potential impacts before exploring joint or explicitly transboundary models.
- Process uncertainty and selectivity (Planned/Partially addressed).
- Estimation of selectivity random-walk variances via marginal likelihood is now feasible in the RTMB “rpm” implementation (and in RCEATTLE/SPOCK), to be explored after the Russian scenarios.
- Time-varying M remains a cautious exploration topic (informed by CEATTLE multi-species work).
- Growth & weight-at-age (Deferred). Cohort and temperature effects on WAA acknowledged; staged for later development.
- Data weighting / likelihoods (Deferred but available). Dirichlet-multinomial (DM) for composition fits (Laplace) is available (RCEATTLE, SPOCK, RTMB “rpm”); bootstrap-tuned effective sample sizes retained for now.

Bridging ADMB $\rightarrow$ RTMB (what changed, what matched) • Minimal code changes to ADMB logic to make it RTMB-compatible: • Fishery dome-shaped penalties applied to log-selectivities for ages >6 . • “Deviation-about-a-mean” parameterizations re-expressed as level-parameters with mean-centered likelihoods (e.g., `log_avgrec`, `log_avg_F`, BTS selectivity slope). • Verification. Using ADMB MLEs in RTMB yields matching NLL components, near-zero gradients, and overlapping SSB/recruit series and asymptotic SDs. Differences are small and explained by the reparameterizations. • Demonstration of efficient MCMC application of ADNUTS package to the RTMB model.

Russian vs. U.S. assessments (EBS–WBS) • Signals match; magnitudes differ. Interannual patterns (especially age-1 recruits) are broadly synchronous; U.S. numbers-at-age are typically $\sim 2\text{--}4\times$ Russian values (peaking $>5\times$ in early 2010s). • Method/management contrasts. • Russia: modern MSY rule with $F_{\text{target}}=F_{\text{MSY}}$ and low B_{trigger} (5th percentile of SSB_{MSY}); recruitment treated as weakly stock-dependent $\rightarrow$ higher TACs when conditions are good, higher risk if recruitment dips. • U.S.: precautionary $F_{\text{ABC}} < F_{\text{MSY}}$, $B_{\text{35\%}}$ proxy, explicit SRR and uncertainty $\rightarrow$ lower volatility and risk. • Implication. Harmonized interpretation needs to account for different HCRs, survey systems, and model structures; scenario inclusion of Russian removals is the immediate step.

EBS vs. GOA comparison (context) • Scale and coupling. EBS abundances exceed GOA by roughly $5\text{--}15\times$ across ages. Some weak–moderate coupling at early ages (shared environmental drivers), little coherence at older ages; populations are largely decoupled.

Technical notes added for transparency • Selectivity penalties. First-difference penalties over time and age are documented with zero-anchor conventions and RTMB R implementations. • Survey likelihood check. Lognormal objective (including constants, on log-scale) provided for parity with ADMB formulations.

Near-term priorities 1. Complete Russian-catch scenarios and quantify sensitivity of status and advice. 2. Switch on/compare marginal-likelihood estimation of selectivity process variances (RW penalties). 3. Pilot DM composition fits and contrast with bootstrap-tuned effective N. 4. Stage cohort/temperature WAA models and scoped exploration of time-varying M (informed by CEATTLE).

Bottom line. The platform migration is done and validated; the biggest outstanding uncertainty for 2025 advice is external removals (Russia). Process-error and likelihood improvements are queued behind that analysis to avoid confounding sources of change.

2 Response to 2025 CIE Review of EBS Walleye Pollock Assessment

This section outlines the response to the findings and recommendations of the 2025 Center for Independent Experts (CIE) review of the Eastern Bering Sea (EBS) walleye pollock stock assessment. The final peer-reviewed reports from the three reviewers are from [Matt Cieri](#), [Yan Jiao](#), and [Anders Nielsen](#).

One of the reviewers' primary concerns was the continued reliance on ADMB. This has now been resolved with the transition to RTMB, which enables use of random-walk selectivity and recruitment deviations, improved error propagation through marginal likelihood, and direct integration of retrospective diagnostics such as Mohn's rho. This task is complete.

Another prominent theme was the absence of Russian removals from the assessment model. All three reviewers emphasized the need to explore the magnitude of this omission, both in terms of sensitivity runs and in better quantifying transboundary exchange. Work is in progress, with planned runs assuming Russian removals of 20–45% of U.S. catch based on (Japp and Sharov (2024)) and reviewer guidance. These scenarios will provide bounds on the possible scale of impacts before moving toward more collaborative or joint modeling approaches.

With respect to process uncertainty, selectivity, and growth, the reviewers identified several areas where additional flexibility could be valuable. Anders recommended estimating process variances for random-walk selectivity penalties using marginal likelihood, rather than fixing them. This aspect is possible within RCEATTLE and SPOCK currently, and is now available in the RTMB version of the pollock model (`rpm`). Yan noted that weight-at-age models could incorporate cohort and temperature effects, and also suggested cautious exploration of time-varying natural mortality, potentially informed by multispecies work such as CEATTLE. These issues have been partially addressed, and further exploration of process error variances will follow after the Russian catch work is integrated.

On data weighting and likelihood specification, Matt Cieri highlighted the potential advantages of a Dirichlet–multinomial (DM) framework estimated via Laplace approximation, as a more realistic representation of composition variance. For now, the bootstrap-based effective sample size approach has been retained, with DM implementation available in three different platforms (RCEATTLE, SPOCK, and the converted RTMB pollock model). We anticipate examining this as an option in the coming year.

Table 2.1: CIE 2025 recommendations and response summary

Theme	Lead Reviewer	Recommendation	Status
RTMB transition	All	Port model from ADMB to RTMB	Complete
Russian catch	All	Incorporate or test Russian removals	In progress
Selectivity complexity	Nielsen	Estimate process variances via marginal likelihood	Planned
Natural mortality	Jiao	Explore M variation and CEATTLE integration	Conceptual
Growth & WAA	Jiao	Model cohort and temperature effects on WAA	Deferred
Data weighting	Cieri	Test Dirichlet-multinomial likelihoods	Available
Retrospective	Nielsen	Use Mohn’s rho over shorter periods	Complete

A consolidated summary of reviewer recommendations and their current status is provided in Table [Table 2.1](#).

Since the review, we adopted the ADMB operational model used in 2024 and ported it to RTMB. We verified that outputs match the between versions. Secondly, we have taken initial steps to address Russian catch uncertainty by comparing of assessments between regions.

3 Bridging ADMB to RTMB

The ADMB code used for the 2024 EBS pollock assessment has been fully ported to RTMB, with verification that outputs match the prior version. This transition enables several important enhancements, including the ability to estimate random-walk selectivity and recruitment deviations, improved error propagation through marginal likelihood, and greater transparency for code review and modifications. The original ADMB code has several options for model configurations, including different selectivity models, data inclusion, and process error structures. Only the base-case 2024 configuration was fully ported over to RTMB with some minor modifications as described below.

We changed the ADMB code as little as possible to enable it to be compatible and work within the RTMB framework. One change was the use of an `if` statement in the penalty for dome-shaped selectivity in the fishery. In the original ADMB code, the penalty was invoked *if* the selectivity at one age was less than the selectivity at the previous adjacent younger age. The modification to this was simply to penalize log-selectivities of adjacent age groups older than age 6.

The ADMB code also has several cases where say y_i is a function of a parameter θ and a deviation ϵ_i such that $y_i = \theta + \epsilon_i$. In ADMB, the deviations are estimated as a vector of parameters with a constraint to have mean zero., In RTMB, it seems best to reparameterize them as a simple vector of parameters (say as θ_i) and then do the likelihood parts as deviations from the mean. To this end, we modified the following parameters which were parameterized as deviations from the mean (e.g., `log_avgrec` (average recruitment), `log_avg_F` (average fishing mortality), `log_slp_bts` (slope of the selectivity for the bottom trawl)) to be simply the vector of values. These changes resulted in a slight change in the pattern of spawning biomass and recruitment estimates among the 2024 ADMB model and the 2024 modified version of the ADMB model used for comparisons with the RTMB results (Figure 1).

Once these minor modifications were made, the RTMB code was able to replicate the results from feeding the MLE parameter estimates from ADMB into the RTMB code. This gave virtually the same estimates of spawning stock biomass (SSB) and age 1 recruits (Figure 2; Figure 3).

```
library(ebswp)
library(patchwork)
pm_orig <- read_rep(here::here("runs", "lastyr", "pm.rep")) # Read in the report file)
pm_mod  <- read_rep(here::here("runs", "rtmb", "pm.rep")) # Read in the report file)
```

```

sd_orig <- read_table(here::here("runs/lastyr", "pm.std")) |> filter(name=="pred_rec" )
sd_mod  <- read_table(here::here("runs/rtmb", "pm.std"))|> filter(name=="pred_rec" )
df <- rbind( data.frame( name = "SSB",
  value = as.numeric(pm_orig$SSB[,2]),
  std.dev = as.numeric(pm_orig$SSB[,3]),
  model = "ADMB original" ) ,
  data.frame( name = "SSB",
  value = as.numeric(pm_mod$SSB[,2]),
  std.dev = as.numeric(pm_mod$SSB[,3]),
  model = "ADMB Modified") )
df <- rbind(df, data.frame( name = "Recruits",
  value = as.numeric(sd_orig$value),
  std.dev = as.numeric(sd_orig$std.dev),
  model = "ADMB original" ) ,
  data.frame( name = "Recruits",
  value = as.numeric(sd_mod$value),
  std.dev = as.numeric(sd_mod$std.dev),
  model = "ADMB Modified") )
df$Year <- rep(1964:2024, 4)

p1 <- df |> filter(name=="SSB",Year>1977) |>
  ggplot(aes(x=Year, y=value, color=model)) + geom_point() + geom_line(aes(group=model)) +
  geom_ribbon(aes(ymin=value-2*std.dev, ymax=value+2*std.dev, fill=model), alpha=0.2, color=
  ggthemes::theme_few() + ylab("SSB (t)") + xlab("Year") +
  scale_y_continuous(labels = scales::comma, limits=c(0,NA)) +
  scale_x_continuous(breaks=seq(1965,2025,5))

p2 <- df |> filter(name=="Recruits", Year>1977) |>
  ggplot(aes(x=as.factor(Year), y=value, color=model)) +
  geom_point(size=2, position=position_dodge(width=0.8)) +
  geom_errorbar(aes(ymin=value-2*std.dev, ymax=value+2*std.dev),
    alpha=0.8, width=0.8, position=position_dodge(width=0.8)) +
  ggthemes::theme_few() +
  ylab("Age 1 Recruitment (thousands)") +
  xlab("Year") +
  scale_y_continuous(labels = scales::comma) +
  expand_limits(y = 0) +
  scale_x_discrete(breaks=seq(1980,2025,5))
p1/p2

```

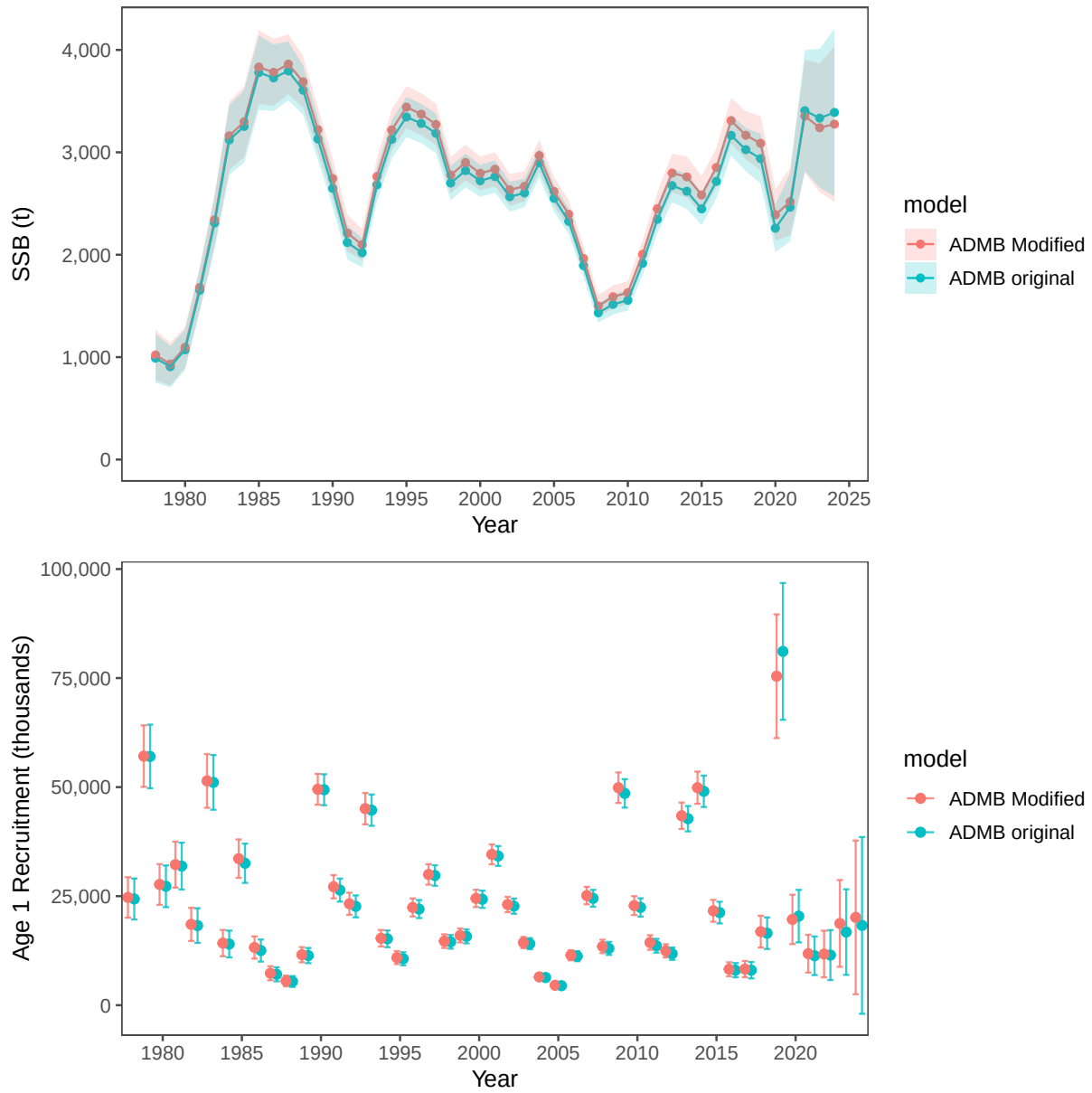


Figure 1: Differences in SSB and recruits between the base ADMB used in 2024 and the modifications for bridging with RTMB. Note that these are slight modifications of the ADMB configurations.

3.1 RTMB Model implementation

The RTMB model code is implemented in the `R/` directory, with the main entry point in `R/Rpm.R`. The initial steps to bridge ADMB to RTMB involve identifying key components and their behaviors. We created a table function that was helpful to evaluate which components of the model in RTMB tracked with the ADMB version. When the bridging aspect was achieved there were de minimus differences between the two model versions (Table 3.1). Additionally, after some initial testing we were able to take the parameter estimates from the ADMB model and show that the gradients were very close to zero and the negative log-likelihood values matched (`tbl-gradient`).

Table 3.1: Comparison of the rtmb code (RPM) and the ADMB code results.

Opening /Users/jim/_mymods/noaa-afsc/EBS_pollock/runs/data/wtage2024.dat
[

Comparison of RTMB and ADMB Outputs

Tolerance = 1e-05] Comparison of RTMB and ADMB Outputs

Tolerance = 1e-05					
Variable	Equal (tol)	Length	Max Abs diff	Max % diff	Correlation
N	TRUE	915	0.049435	0.000485	1.000000
Z	TRUE	915	0.000000	0.000159	1.000000
F	TRUE	915	0.000000	0.000477	1.000000
M	TRUE	915	0.000000	0.000000	1.000000
S	TRUE	915	0.000000	0.000122	1.000000
pred_catch	TRUE	61	0.004932	0.000380	1.000000
obs_catch	TRUE	61	0.005000	0.000458	1.000000
SSB	TRUE	61	0.004977	0.000416	1.000000
phizero	TRUE	1	0.000000	0.000189	NA
Bzero	TRUE	1	0.003401	0.000056	NA
steepness	TRUE	1	0.000000	0.000070	NA
pred_cpue	TRUE	12	0.004213	0.000120	1.000000
pred_avo	TRUE	18	0.000005	0.000413	1.000000
eb_bts	TRUE	42	0.048608	0.000455	1.000000
eb_ats	TRUE	19	0.004484	0.000223	1.000000
sel_fsh	TRUE	915	0.000005	0.000491	1.000000
sel_bts	TRUE	645	0.000000	0.000405	1.000000
sel_ats	TRUE	465	0.000005	0.000466	1.000000
sam_fsh	TRUE	60	0.000499	0.000134	1.000000
sam_bts	TRUE	42	0.000000	0.000000	1.000000
sam_ats	TRUE	19	0.000000	0.000000	1.000000
phat_fsh	TRUE	900	0.000000	0.000464	1.000000
phat_bts	TRUE	630	0.000000	0.000471	1.000000
phat_ats	TRUE	285	0.004822	0.000436	1.000000
age_like	TRUE	3	0.001044	0.000200	1.000000
age_like_offset	TRUE	3	0.039496	0.000326	1.000000
cat_like	TRUE	1	0.000002	0.000084	NA
bts_like	TRUE	1	0.000014	0.000041	NA
ats_like	TRUE	1	0.000000	0.000003	NA
ats_age1_like	TRUE	1	0.000025	0.000220	NA
cpue_like	TRUE	1	0.000002	0.000190	NA
avo_like	TRUE	1	0.000005	0.000055	NA
avgsel_like	TRUE	1	0.000000	0.000207	NA
wt_nll	TRUE	4	0.004116	0.000172	1.000000
wt_like	TRUE	1	0.004805	0.000076	NA
rec_like	TRUE	7	0.000040	0.000187	1.000000
Fpen_like	TRUE	1	0.000002	0.000026	NA
sel_like	TRUE	3	0.000021	0.000124	1.000000
sel_like_dev	TRUE	3	0.000465	0.000375	1.000000
Priors	TRUE	4	0.000025	0.000126	1.000000
tot_like	TRUE	1	0.004098	0.000052	NA

EBS_pollock assessment discussion paper

Table 3.2: Top 20 parameters with largest gradients (absolute value) from RTMB using ADMB parameter estimates.

outer mgc: 6.168831e-05 outer mgc: 6.168831e-05	
Parameter	gradient
log_q_cpue	-6.168831e-05
log_F_devs	6.048117e-05
sel_devs_atc	-4.178819e-05
log_F_devs	-3.994676e-05
log_F_devs	-3.965254e-05
sel_devs_atc	-3.679200e-05
log_rec_devs	3.621454e-05
log_F_devs	3.543620e-05
log_F_devs	3.507003e-05
log_F_devs	-3.475311e-05
log_F_devs	-3.454961e-05
sel_devs_atc	-3.219652e-05
log_F_devs	-3.198028e-05
log_F_devs	2.904682e-05
log_F_devs	2.786280e-05
sel_a50_bts_dev	2.688229e-05
log_rec_devs	-2.644907e-05
log_F_devs	-2.617959e-05
log_F_devs	-2.492655e-05
log_F_devs	2.483619e-05

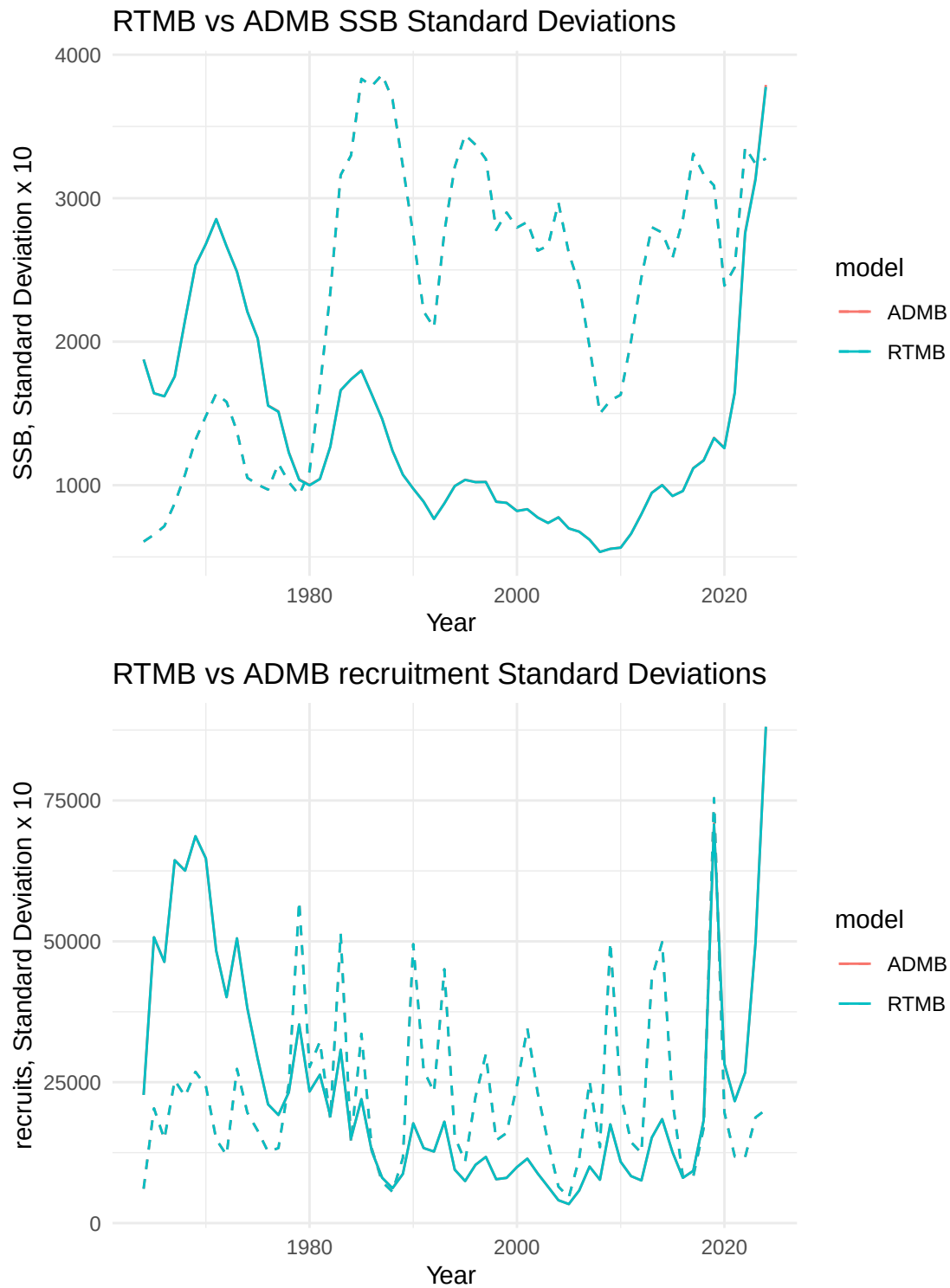


Figure 2: SSB and age-1 recruits from RTMB and ADMB, dashed lines are standard deviations (x10). Both ADMB and RTMB results are overlain.

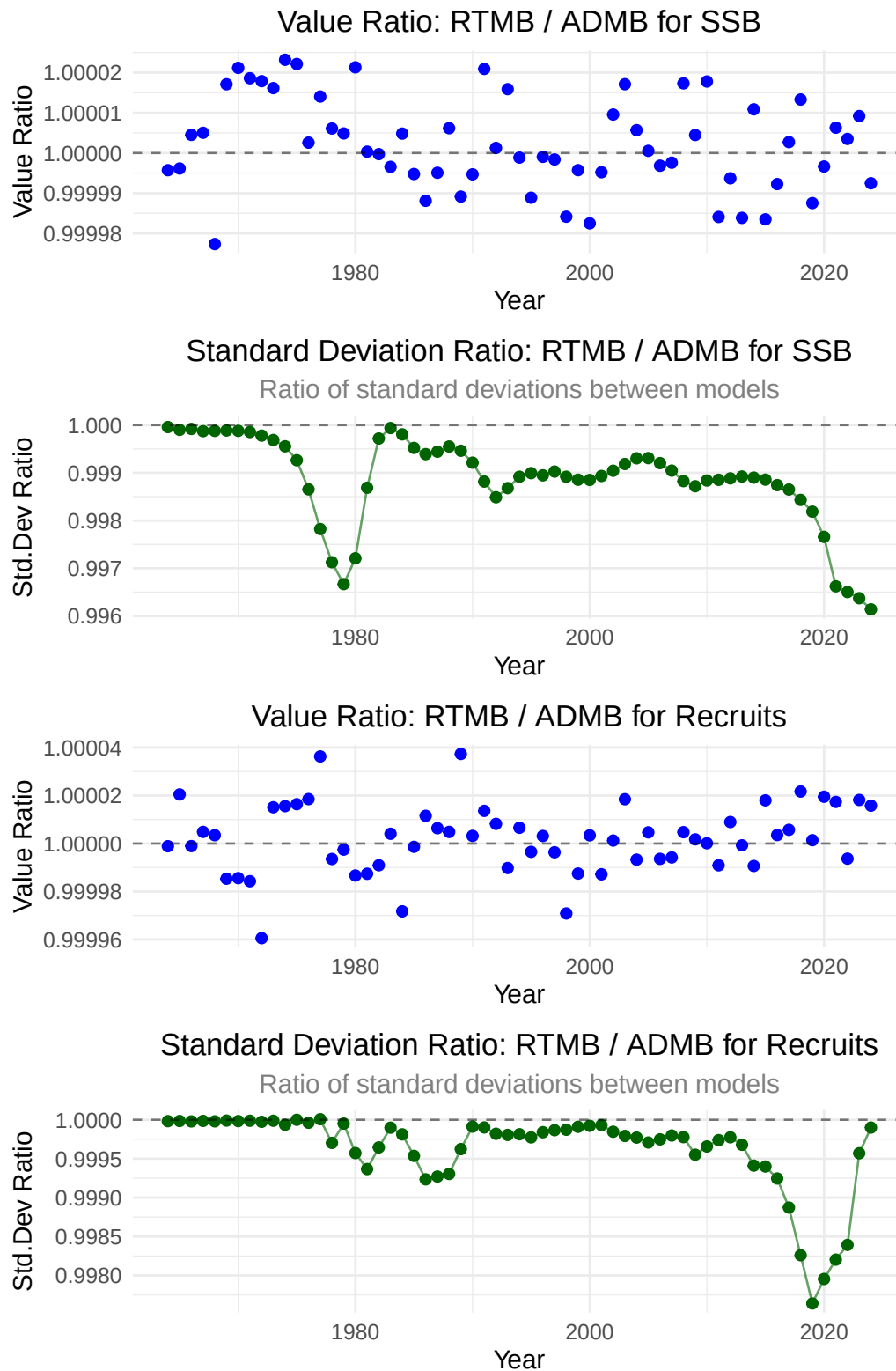


Figure 3: Relative differences between ADMB and RTMB estimates of SSB and age-1 recruits and asymptotic standard deviation approximations.

4 Considerations for future assessments regarding neighboring Russian federation fisheries

As this year’s assessment moves forward, particular attention is warranted on how Russian fisheries and assessments are considered alongside our own results. Pollock in the eastern and western Bering Sea are generally recognized as a single biological stock, with seasonal exchange across the U.S.–Russia boundary documented by acoustic studies. As noted in the CIE reviews, Russian removals are substantial with recent estimates representing on the order of 20–45% of the U.S. catch in some years.

While examined in past assessments, the Russian catch magnitude should be re-evaluated to examine how they might alter management advice. As a step in beginning to make a more detailed evaluation, we compared the Russian stock assessment results presented as part of their Marine Stewardship Council (MSC) review panel with our model results (through 2024). The following sections compare outcomes from the assessments presented to

4.1 Comparison of Russian and USA age-structured assessment results

4.1.1 Comparison of Russian WBS and U.S. EBS Pollock Assessments

Feature	Russian WBS Assessment	U.S. EBS Assessment
Primary model	TISVPA catch-at-age model (Vasilyev 2005); Synthesis model retained as supporting tool	ADMB statistical catch-at-age model (pollock model), routinely updated
Assessment frequency	Annual assessment; fishery-independent surveys ~ every 2 years	Annual assessment; multiple fishery-independent surveys each year (BTS, ATS, AVO)
Reference points (HCR)	Since 2022: $F_{target} = F_{msy}$ (or proxy); $B_{trigger} = 5th$ percentile of SSB_{msy} distribution	Precautionary: $F_{ABC} < F_{msy}$, $B_{35\%}$ proxy for SSB_{msy} ; reference points set conservatively

Feature	Russian WBS Assessment	U.S. EBS Assessment
Management philosophy	“Modern MSY” approach, aiming for maximum sustainable harvest without exceeding safe stock status; allows ~15–30% higher TACs vs. precautionary approach	Precautionary approach emphasizes risk reduction; TAC constrained by Council harvest rules
Recruitment assumption	Recruitment assumed independent of stock size, fluctuating randomly within historical distribution	Stock–recruit relationship incorporated (steepness parameter), with uncertainty reflected in projections
Recent stock status	2023: SSB well above SSB _{target} and SSB _{msy} proxy; $F < F_{\text{target}}$ and F_{limit} ; stock increasing	2024: SSB above B35% and B _{msy} proxy; F well below reference points; stock stable to increasing
Uncertainty notes	Higher terminal year uncertainty in TISVPA compared to Synthesis; risk of overexploitation if stock declines under new HCR	Uncertainty routinely quantified (likelihood profiles, MCMC); precautionary buffers applied in catch advice
Governance & compliance	Managed by Federal Fisheries Agency (FFA) and CFMC; MSC certified since 2021; electronic logbook system in place	Managed under NPFMC/AFSC framework; extensive observer coverage; SAFE reports reviewed annually

The contrasts between the Russian WBS and U.S. EBS pollock assessments highlight both complementary perspectives and important methodological differences. Both systems conclude that pollock biomass is currently above target reference levels and that fishing mortality is below limits, reinforcing the perception of a healthy stock across the Bering Sea. However, the Russian shift to a “modern MSY” framework, with F_{target} set at F_{msy} and B_{trigger} at the lower 5th percentile of SSB_{msy}, allows higher catch levels but introduces greater risk if recruitment declines. By contrast, the U.S. precautionary approach constrains harvest below F_{msy} and maintains conservative biomass thresholds, prioritizing long-term risk reduction. These differences underscore the need for careful interpretation when comparing results across assessments: while the overall trends align, differences in model structure, reference points, and treatment of uncertainty can produce divergent management advice even under similar stock conditions.

4.1.2 Russian literature context: MSY vs. Precaution in WBS pollock

Bulatov & Vasilyev (2023) argue for moving from a precautionary HCR to a modern *MSY-based rule* for WBS/Navarin and N. Okhotsk pollock. Their stochastic projections (using *TISVPA*)

resample historical recruitment (assumed independent of SSB) and set $F_{target} \geq F_{MSY}$, with MSY $B_{trigger}$ taken as the 5th percentile of the SSB distribution under F_{MSY} . Reported values include:

- F_{MSY} 0.75 yr⁻¹ (Navarin, WBS) and **0.60 yr⁻¹** (N. Okhotsk).
- $B_{trigger}$ 470 kt (Navarin) and **1.77 Mt** (N. Okhotsk).
- Expected catches **15–30% higher** than under the precautionary rule (2002–2022 reconstructions), with large short-term gains in 2022 (e.g., Navarin TAC +~60%).

Implications for our EBS–WBS comparison: the Russian MSY framing aligns with the recent WBS HCR updates ($F_{target}=F_{MSY}$, low B-trigger), helping explain higher Russian TAC recommendations when recruitment is strong. The approach is intentionally less conservative and hinges on **weak SRR signal** assumptions; if recruitment dips, risk tolerance differs from the EBS precautionary setup. We’ll reference this when interpreting USA–Russia abundance ratios and management advice.

Datsky, Sheybak & Antonov (2022)** provide a comprehensive synthesis of pollock distribution, biology, stock status, and fishery dynamics across the Bering Sea. Drawing on >25 years of Russian survey and fishery data, they describe the **ecological plasticity** of pollock, its broad bathymetric range (20–600 m), and the role of **seasonal migrations** linking eastern and western Bering Sea populations.

Key points include:

- **Population structure:** major groupings (eastern Bering Sea, Anadyr–Navarin, western Bering Sea) have distinct spawning centers but mix during feeding migrations, particularly in the northwestern Bering Sea.

- **Stock trends:** the western Bering Sea stock declined to lows in the late 1990s–early 2000s, but has shown **gradual recovery** since 2010. East Bering Sea and NW Bering Sea biomasses are currently at average long-term levels, with fluctuations strongly influenced by climate, ice, and prey availability.
- **Fishery importance:** pollock accounts for ~70% of catches in Russian Bering Sea waters, dominated by midwater trawls, with recent landings of ~350–630 kt/yr. Seasonal patterns show ~90% of catches taken June–December.

Implications for U.S.–Russia comparisons: This work emphasizes that western and northwestern pollock are not isolated; they are reinforced by migrations from the eastern Bering Sea, meaning observed differences in assessment outputs (biomass, recruitment) partly reflect shared dynamics across the basin as well as differences in survey intensity and assessment methods.

Table 4.2: Correlations between Russian and USA age classes, 1995-2024.

[

Correlations between Russian and USA Age Classes] Correlations between Russian
and USA Age Classes

Age	Correlation
1	0.625
2	0.627
3	0.618
4	0.697
5	0.682
6	0.560
7	0.083
8	0.098
9	0.007
10	-0.500

Table 4.3: Summary statistics of magnitude differences between Russian and USA age classes, 1995-2024.

[1] “mean ratio (USA/Russian): 2.87” [1] “median ratio (USA/Russian): 2.7”

Pollock 99% of catch; seabird & marine mammal mortality very low Artyukhin, Korobov, and Gluschenko (2023); Burdin (2021)]. (Japp and Sharov 2023, 2024; Bulatov and Vasilev 2023). Datsky, Sheybak, and Antonov (2022),

The comparison of U.S. and Russian pollock assessments shows that U.S. estimates of total numbers-at-age are consistently higher than those from Russia, typically by a factor of 2–4, with the ratio peaking above 5 in the early 2010s (Figure 4, bottom-left). Both assessments show similar interannual patterns, particularly for age-1 recruits (top-right), suggesting some shared signals of strong year classes. Correlation analysis by age class (bottom-right; Table 4.2) confirms this alignment, with moderate to strong positive correlations across ages 1–6. At older ages, correlations weaken and even turn negative (age 10), reflecting differences in how the two assessments treat age structure and fishery/survey data inputs. Overall, while the magnitude of estimates differs substantially between the two countries, the consistent synchrony in recruitment trends indicates that both assessments are capturing the same broad population dynamics, albeit with systematic differences in scale.

=== DETAILED STATISTICS ===

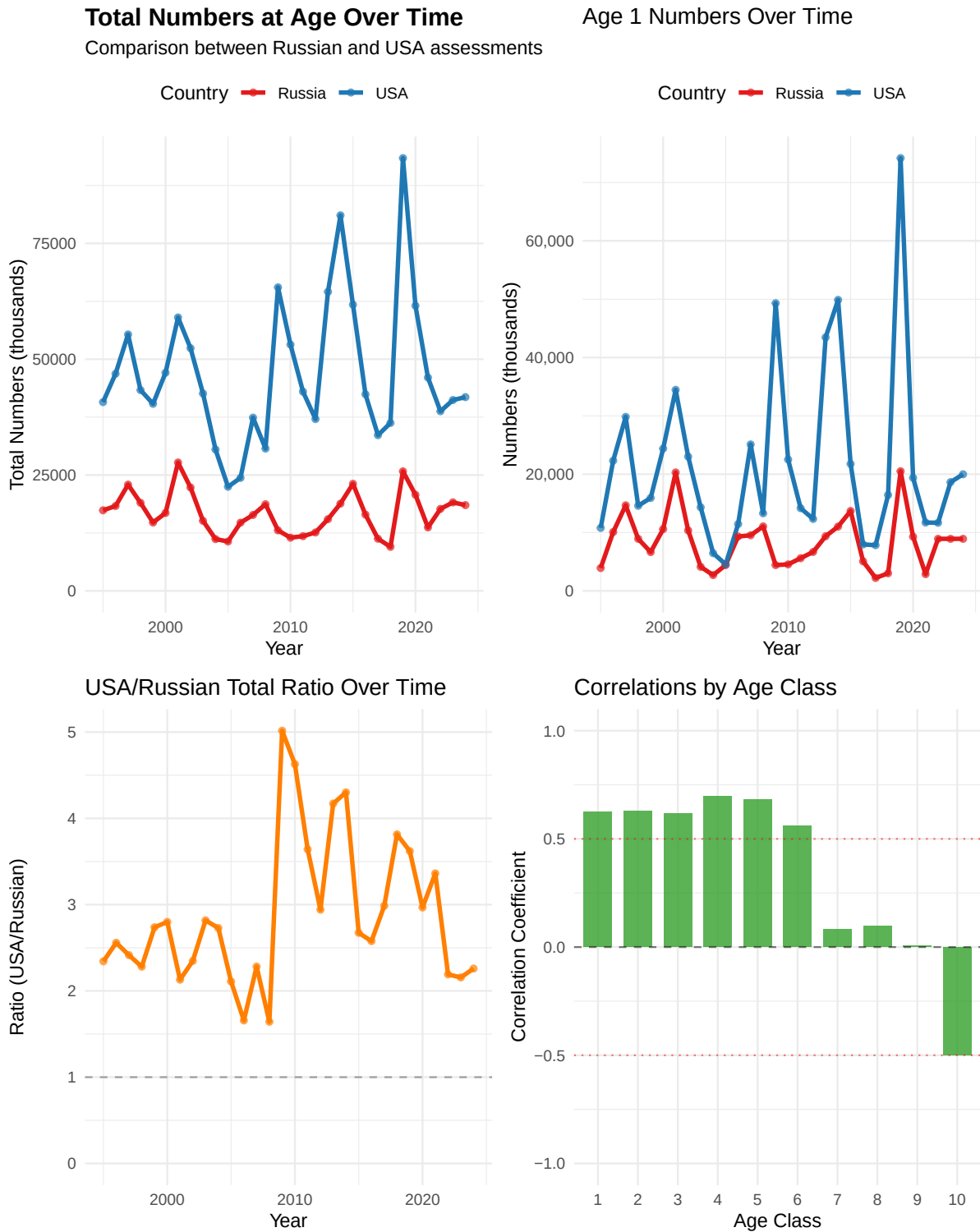


Figure 4: Various comparisons between Russian and USA age-structured survey results.

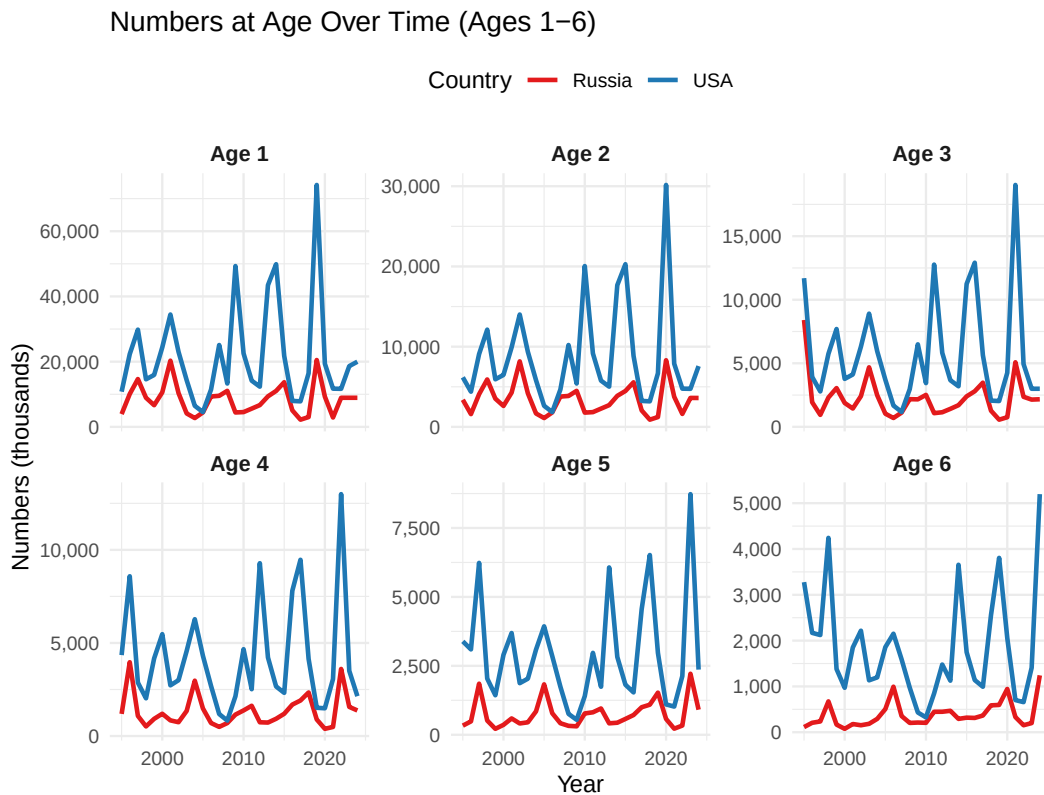
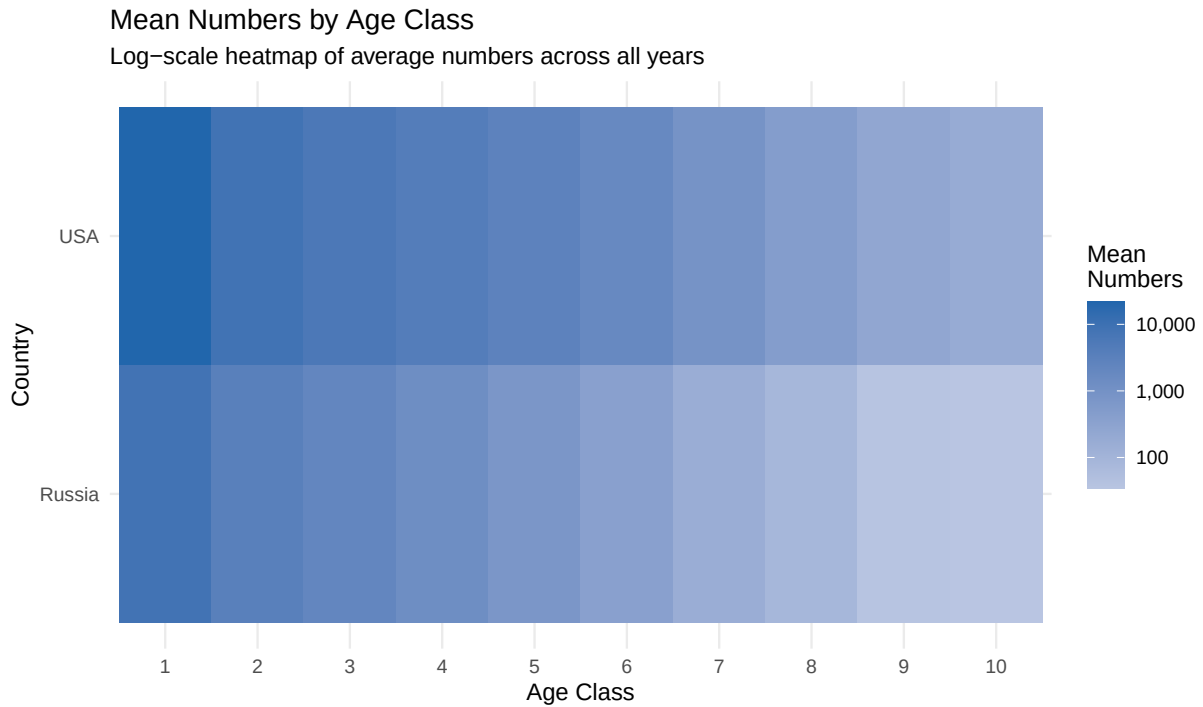


Figure 5: Various comparisons between Russian and USA age-structured survey results.

Summary statistics by country and age:

A tibble: 20 x 8

	Country	Age	Mean	Median	SD	CV	Min	Max
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	Russia	1	8391.	8925	4647.	0.554	2213.	20509.
2	Russia	2	3400.	3562.	1860	0.547	900.	8308.
3	Russia	3	2254.	2154.	1574.	0.699	565.	8406.
4	Russia	4	1338.	1169.	875	0.654	387.	3966.
5	Russia	5	743	575.	511.	0.688	216.	2214.
6	Russia	6	382.	302.	278	0.728	74.5	1242.
7	Russia	7	168.	138.	127.	0.757	15.3	550.
8	Russia	8	89.3	65.2	81.9	0.917	0.8	367.
9	Russia	9	38.7	31.2	37.3	0.964	0.4	178.
10	Russia	10	35.1	16.8	45.4	1.29	0.2	164.
11	USA	1	21725.	17539.	15273.	0.703	4546.	74180.
12	USA	2	8768.	6580.	6227.	0.71	1848.	30158.
13	USA	3	5797.	4186	4098.	0.707	1175.	19030.
14	USA	4	4232.	3283.	2852.	0.674	831.	12994.
15	USA	5	2912.	2594.	1889.	0.648	543.	8729.
16	USA	6	1842.	1535.	1174.	0.637	325	5198.
17	USA	7	935.	809.	565.	0.604	185.	2560.
18	USA	8	488	375.	317.	0.649	88.1	1532
19	USA	9	260.	204.	178.	0.685	41.8	861.
20	USA	10	188	176.	103.	0.547	37.8	476.

Coefficient of Variation by age:

A tibble: 10 x 3

	Age	Russia_CV	USA_CV
	<dbl>	<dbl>	<dbl>
1	1	0.554	0.703
2	2	0.547	0.71
3	3	0.699	0.707
4	4	0.654	0.674
5	5	0.688	0.648
6	6	0.728	0.637
7	7	0.757	0.604
8	8	0.917	0.649
9	9	0.964	0.685
10	10	1.29	0.547

Age classes with correlation > 0.5 : Age1, Age2, Age3, Age4, Age5, Age6

Age classes with correlation < 0.2 : Age7, Age8, Age9, Age10

Age1	Age2	Age3	Age4	Age5	Age6	Age7	Age8	Age9	Age10
0.625	0.627	0.618	0.697	0.682	0.560	0.083	0.098	0.007	-0.500

4.2 Results comparing Russian and USA age-structured assessments

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4.3 Comparison with the Gulf of Alaska (GOA) pollock estimates

Since the Russian context suggest some connectivity, for perspective, we thought it might be useful to conduct a similar comparison with results from the GOA pollock stock. Pollock assessments from the eastern Bering Sea (EBS) and Gulf of Alaska (GOA) show strong contrasts in both magnitude and dynamics of numbers-at-age. The EBS consistently supports far higher abundances across all ages and years (typically 5–15 times greater than the GOA; Figure 6). Recruitment pulses are evident in both regions, particularly at age 1, but the correlations by age class reveal only modest positive associations at younger ages, while older ages show little to no correspondence (Table 4.4). This indicates that early life stages may share some environmental drivers across basins, but overall population dynamics remain largely decoupled.

A more detailed breakdown by age class underscores these findings (Figure 7). Cohort strength in the EBS exhibits large variability with recurring strong year classes, while the GOA maintains lower overall abundance with weaker pulses. Time-series plots for ages 1–6 highlight this contrast, with the EBS showing sharp recruitment spikes and the GOA remaining more muted.

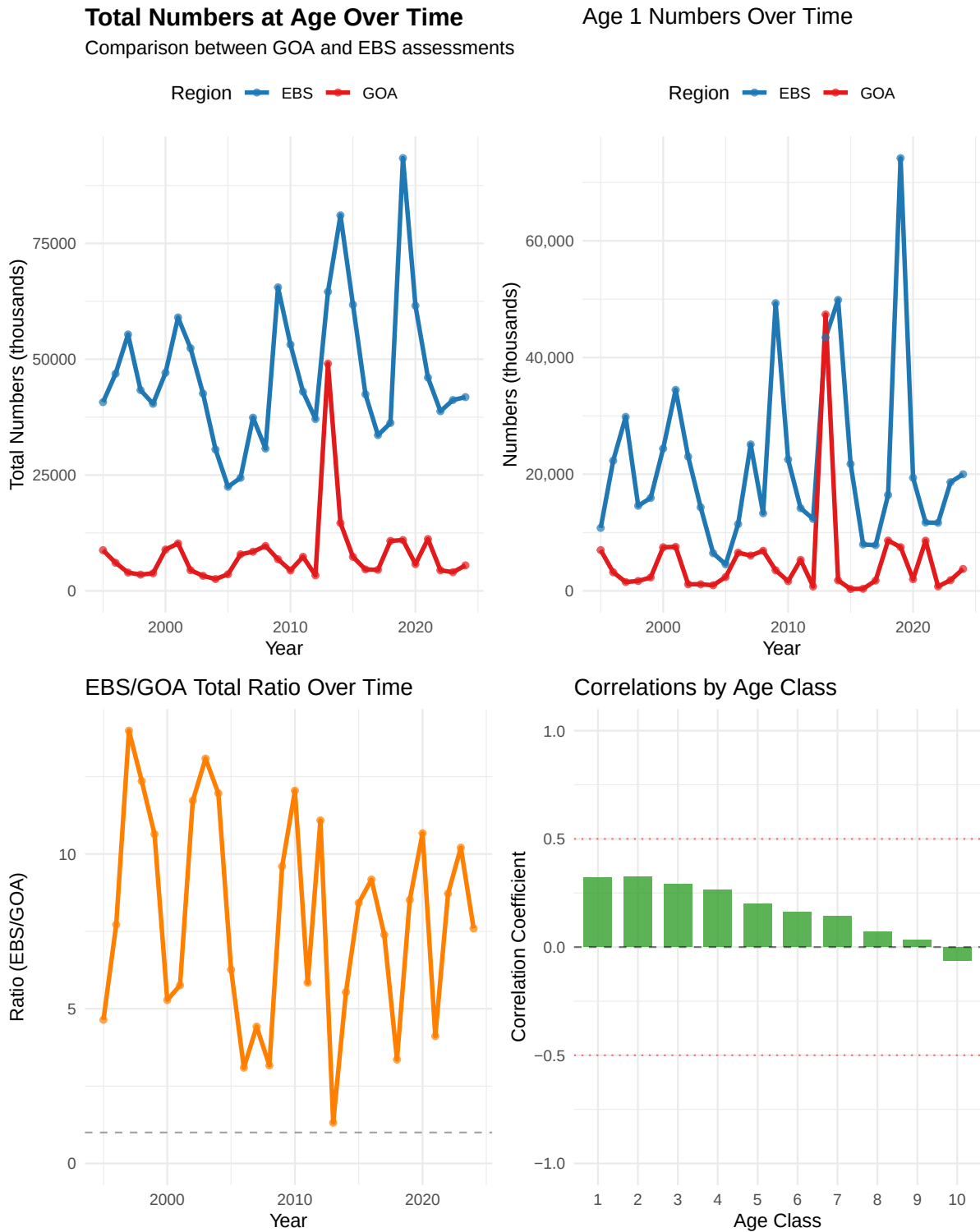
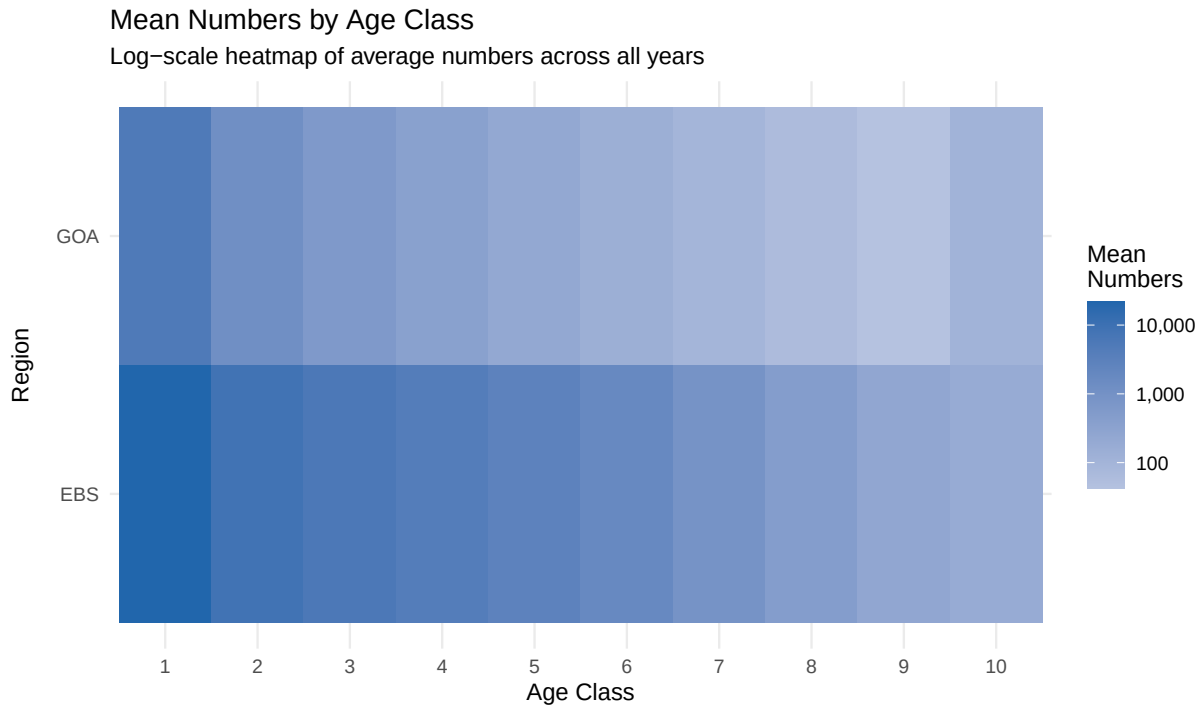


Figure 6: Various comparisons between GOA and EBS age-structured assessment results.



Numbers at Age Over Time (Ages 1–6)

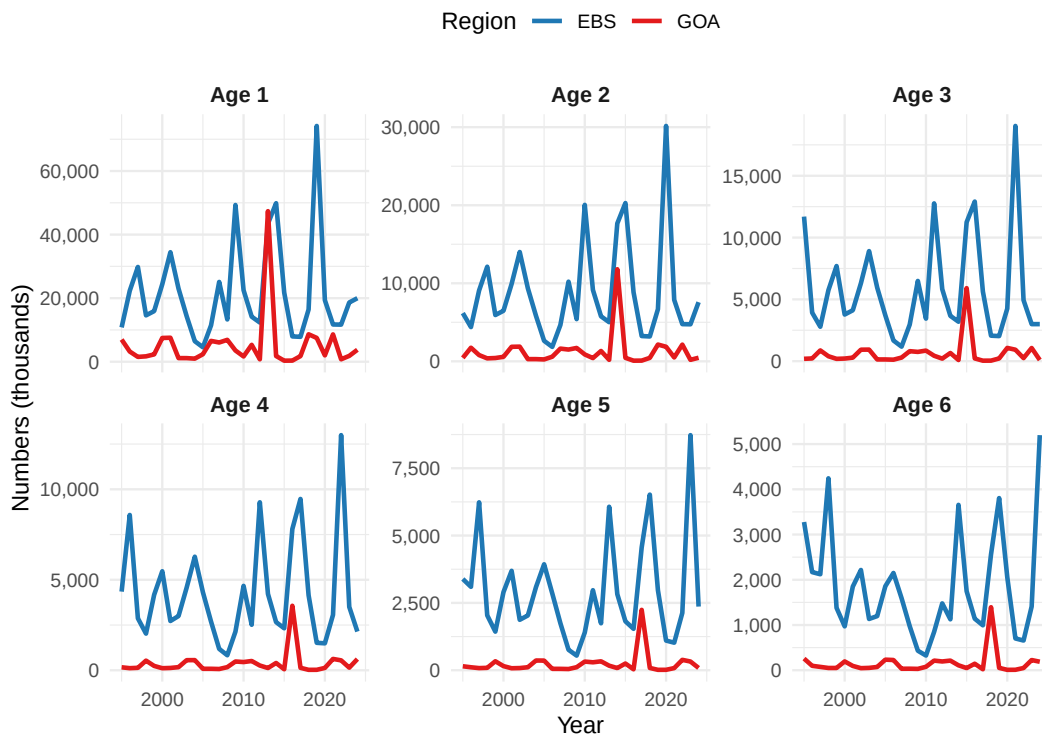


Figure 7: Various comparisons between GOA and EBS age-structured assessment results.

Table 4.4: Summary of age-specific correlations between GOA and EBS age classes, 1995-2024.

[

Correlations between GOA and EBS Age Classes] Correlations between GOA and
EBS Age Classes

Age	Correlation
1	0.321
2	0.324
3	0.289
4	0.264
5	0.200
6	0.162
7	0.142
8	0.071
9	0.032
10	-0.064

5 Other items

The CIE noted in particular that some documentation of aspects of the ADMB code was incomplete. During the meetings, we began to more properly specify how the selectivity penalties were being calculated and implemented.

5.0.1 First-difference selectivity penalty (temporal)

We need the penalty used in ADMB where the **first year is compared to zero** (not to a previous estimated year). For a matrix of log-selectivities ($\log s_{y,a}$) (years ($y = 1, \dots, T$), ages ($a = 1, \dots, A$), the penalty is:

$$\mathcal{P}_{\text{FD,time}} = w \sum_{a=1}^A \left[(\log s_{1,a} - 0)^2 + \sum_{y=2}^T (\log s_{y,a} - \log s_{y-1,a})^2 \right].$$

RTMB/R implementation

```
# log_sel: matrix [T x A] of log-selectivities (year rows, age cols)
# w: scalar or length-A vector of weights
fd_penalty_time <- function(log_sel, w = 1) {
  stopifnot(is.matrix(log_sel))
  T <- nrow(log_sel); A <- ncol(log_sel)
  if (length(w) == 1) w <- rep(w, A)
  stopifnot(length(w) == A)

  diffs <- rbind(log_sel[1, , drop = FALSE] - 0,      # year 1 vs 0
                 apply(log_sel, 2, diff))              # y>=2 diffs
  sum(colSums((sqrt(w) * diffs)^2))
}
```

5.0.2 Sanity check

```
set.seed(42)
T <- 5; A <- 3
L <- matrix(rnorm(T*A, sd=0.2), T, A); w <- c(1, 2, 0.5)

fd_loop <- function(L, w){
  acc <- 0
  for (a in seq_len(ncol(L))) {
    acc <- acc + w[a]*(L[1,a]-0)^2
    for (y in 2:nrow(L)) acc <- acc + w[a]*(L[y,a]-L[y-1,a])^2
  }
  acc
}

all.equal(fd_penalty_time(L, w), fd_loop(L, w))
```

```
[1] "Mean relative difference: 0.01953004"
```

5.0.3 First-difference selectivity penalty (age)

For within-year smoothing over age with a **zero anchor at the minimum age**:

$$\mathcal{P}_{\text{FD,age}} = w \sum_{y=1}^T \left[(\log s_{y,a_{\min}} - 0)^2 + \sum_{a=a_{\min}+1}^A (\log s_{y,a} - \log s_{y,a-1})^2 \right].$$

RTMB/R implementation

```
# log_sel: matrix [T x A] of log-selectivities (year rows, age cols)
# a_min: column index of minimum age within 'log_sel' (default = 1)
fd_penalty_age <- function(log_sel, w = 1, a_min = 1) {
  stopifnot(is.matrix(log_sel))
  T <- nrow(log_sel); A <- ncol(log_sel)
  stopifnot(a_min >= 1, a_min <= A)
  if (length(w) == 1) w <- rep(w, T)
  stopifnot(length(w) == T)

  # Build diffs across age with anchor at a_min (assumes a_min=1; if not, set zeros below a_min)
```

```

diffs <- matrix(0, nrow = T, ncol = A)
diffs[, a_min] <- log_sel[, a_min] - 0
if (a_min < A) {
  for (a in (a_min+1):A) {
    diffs[, a] <- log_sel[, a] - log_sel[, a-1]
  }
}
sum(rowSums((sqrt(w) * diffs)^2))
}

```

5.1 Reference: survey lognormal scaling (consistency check)

When porting lognormal survey likelihoods, confirm whether ADMB's objective includes the normalizing constant and whether () is on the **log** scale:

$$\ell(\sigma) = -\sum_i \frac{(\log O_i - \log E_i)^2}{2\sigma^2} - n \log \sigma - \frac{n}{2} \log(2\pi).$$

```

lognorm_nll <- function(obs, exp, sigma_log, include_const = TRUE) {
  r <- log(pmax(obs, .Machine$double.eps)) - log(pmax(exp, .Machine$double.eps))
  n <- length(r)
  nll <- sum(r^2) / (2 * sigma_log^2)
  if (include_const) nll <- nll + n*log(sigma_log) + n*0.5*log(2*pi)
  nll
}

```

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